

Inverse Kinematics: Solving for Wrist Angles

~~EGR 445~~

EGR 445/545

Plan for today

- review of wrist location for industrial inverse kinematics
- solving for wrist angles
- quiz 3
- project time?

← last chap. 4 topic

Inverse Kinematics: Big Idea

- inverse kinematics is hard
- find ways to break things into smaller pieces
 - top view
 - side view
 - wrist location

← typically in $\hat{x}_1 / \hat{z}_1$ plane
→ (almost) never in frame 0
↑
?

Inverse Kinematics “Algorithm”

- find ${}^0P_{wrist}$
 - if we have enough information
- draw a top view
 - what variables can we solve for?
- draw a side view
 - what frame do we need to draw the side view in?
 - apply geometric approach to the side view
- verify wrist location based on side view solution
- solve for the wrist angles
- verify final answer
 - reproduce ${}^0T_{tip}$ based on forward kinematics

Typically gives θ_1

Typically gives θ_2 and θ_3

verify that $\theta_1, \theta_2, \theta_3$ are correct

Typically solving for $\theta_4 - \theta_6$

Today

Wrist Location: Big Ideas

- if we can find the wrist/base of the last link, we can make our lives easier by breaking the problem in half
 - this might be the only way to make the problem solvable
- what information do we need to find the base of the last link?

${}^0\mathbf{P}_{w2t}$

- how do we find the wrist-to-tip vector ${}^0\mathbf{P}_{w2t}$ in frame 0?
- ${}^0\mathbf{P}_{w2t} = {}^0\mathbf{P}_{tip} - {}^0\mathbf{P}_{wrist}$
- this is the relative displacement from O_N to the tip
 - the length should be the same in any frame
- typically:

$${}^N\mathbf{P}_{w2t} = \begin{bmatrix} a_{N-1} \\ 0 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 0 \\ 0 \\ d_N \end{bmatrix}$$

DH
where are
the wrist
and tip in
frame N

${}^0\mathbf{P}_{w2t}$ continued

Having ${}^N\mathbf{P}_{w2t}$, how do we find ${}^0\mathbf{P}_{w2t}$?

- this is a relative displacement, so we just need to rotate it
 - we need ${}^0_N\mathbf{R}$, not ${}^0_N\mathbf{T}$
- how do we find ${}^0_N\mathbf{R}$?

From the problem statement

$${}^0_{tip}\mathbf{T} = \begin{bmatrix} {}^0_N\mathbf{R} & {}^0\mathbf{P}_{tip} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- ${}^0_N\mathbf{R} = {}^0_{tip}\mathbf{T} [0:3, 0:3] \leftarrow$
- ${}^0\mathbf{P}_{w2t} = {}^0_N\mathbf{R} {}^N\mathbf{P}_{w2t}$

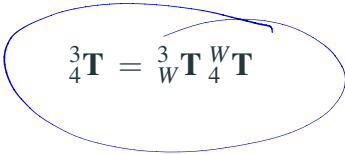
$${}^0\mathbf{P}_{wrist} = {}^0\mathbf{P}_{Tip} - {}^0\mathbf{P}_{w2t}$$

Where is the wrist?

Wrist Matrices

Problem: we are trying to find the wrist, but we don't know θ_4 yet

- the wrist is typically at O_4
- how can I use HT matrices to find O_4 without knowing θ_4 ?


$${}^3_4\mathbf{T} = {}^3_W\mathbf{T} {}^W_4\mathbf{T}$$

$${}^0_{tip}\mathbf{T} = {}^0_1\mathbf{T} \cdot {}^1_2\mathbf{T} \cdot {}^2_3\mathbf{T} \cdot \underbrace{{}^3_W\mathbf{T} \cdot {}^W_4\mathbf{T}} \cdot {}^4_5\mathbf{T}$$

Wrist Matrices Continued

Constants

- ${}^3_W\mathbf{T} = {}^3_4\mathbf{T}|_{\theta_4=0} = \text{DH}(\alpha_3, a_3, 0, d_4)$

- ${}^W_4\mathbf{T} = {}^3_4\mathbf{T}|_{\alpha_3=a_3=d_4=0} = \text{DH}(0, 0, \theta_4, 0)$

} implies d_4 translation
before θ_4 rotation

$${}^3_4\mathbf{T} = {}^3_W\mathbf{T} \cdot {}^W_4\mathbf{T}$$

→ this is ok

only

because

d_4 and θ_4 introduce
the same axis

Solving for Wrist Angles

Solving for Wrist Angles

Big Idea:

$$\overset{\textcircled{1}}{W_{tip} \mathbf{T}_{\text{numeric}}} = \overset{\textcircled{2}}{W_{tip} \mathbf{T}_{\text{symbolic}}}$$

- Find $W_{tip} \mathbf{T}_{\text{symbolic}}$ based on symbolic forward kinematics
 - use `robotics_sympy` and the DH table
- $W_{tip} \mathbf{T}_{\text{numeric}}$ should be all hard numbers
- we have 12 equations (4x4 matrices, but bottom row is [0,0,0,1])
 - pick options that lead to easy equations for θ_4 , θ_5 , and θ_6

← fairly straight forward

Numeric Approach for ${}^W_{tip}\mathbf{T}$

- Find wrist location in frame $\{0\}$
- Find θ_1 from the top view
- Find θ_2 and θ_3 from the wrist position in frame $\{1\}$ using the geometric approach
- Once we have θ_1 , θ_2 , and θ_3 :

$\theta_1, \theta_2, \theta_3 \rightarrow {}^0_W\mathbf{T}_{\text{numeric}} = {}^0_1\mathbf{T} \cdot {}^1_2\mathbf{T} \cdot {}^2_3\mathbf{T} \cdot {}^3_W\mathbf{T}$

${}^0_W\mathbf{T}^{-1} \cdot \left[{}^0_{tip}\mathbf{T} = {}^0_W\mathbf{T} \cdot {}^W_{tip}\mathbf{T} \right]$

${}^W_{tip}\mathbf{T}_{\text{numeric}} = {}^0_W\mathbf{T}^{-1} \cdot {}^0_{tip}\mathbf{T}$

from Problem statement

Quick Review: Finding the wrist in frame 0

$${}^0P_{wrist} = {}^0P_{tip} - {}^0P_{w2t}$$

- what code do we need to find ${}^0P_{wrist}$?

Quick Review: Side View in Frame 1

- what does this mean?
- how do I find ${}^1P_{tip}$ or ${}^1P_{wrist}$?

Fig. 3.15 applied to the wrist

$${}^3_4\mathbf{T} = {}^3_W\mathbf{T} \cdot {}^W_4\mathbf{T}$$

Solving for wrist angles

$${}^W_{tip}\mathbf{T}_{numeric} = {}^W_{tip}\mathbf{T}_{symbolic}$$

- how do we find ${}^W_{tip}\mathbf{T}_{numeric}$?
- how do we find ${}^W_{tip}\mathbf{T}_{symbolic}$?

Simple Example: 3 Link Planar Robot

- using this approach on the robot from LA 9

$$\cdot {}^w T_{\text{Tip}} = {}^w T_{\text{Tip}} S_{g_m}$$

$$\cdot \text{wrist is at } O_3$$

$$\cdot {}^z T_3 = {}^z T_w \cdot {}^w T_3$$

0T symbolic

TIP

i	α	a	θ	d
1	0	0	θ_1	0
2	0	L_1	θ_2	0
3	0	L_2	θ_3	0

$${}^0T = {}^0T_1 \cdot {}^1T_2 \cdot {}^2T_3$$

$${}^0T_{TIP} = {}^0T_1 \cdot {}^1T_2 \cdot {}^2T_3 \cdot {}^3T_{TIP}$$

$${}^0T_{TIP} = {}^0T_3 \cdot {}^3T_{TIP}$$

DH + robotics -
groupy

Inverse Kin. for a “Typical” Industrial Robot

- PUMA, Fanuc, simplified PUMA, ...
- key characteristics:
 - last 2-3 $\hat{Z}$ axes intersect at the wrist
 - wrist moves in a plane set by θ_1
- approach:
 - find ${}^0P_{wrist}$
 - find θ_1 from top view of wrist location
 - find θ_2 and θ_3 from geometric approach on side view
 - probably in frame 1
 - verify wrist location: ${}^0_W\mathbf{T}$
 - find wrist angles from ${}^W_{tip}\mathbf{T}_{numeric} = {}^W_{tip}\mathbf{T}_{symbolic}$
 - verify full solution using forward kinematics

PUMA Wrist Angle Example

- `PUMA_inverse_kinematics_wrist_example.ipynb`
 - in the class 15 folder on github